15 (a) Find the value of $\sqrt[4]{27 \times 3 \times 10^8}$

$27 \times 3 \times 10^8 = 81 \times 10^8 = 3^4 \times 10^8.$

1 mark
Fourth root of 3^4

Fourth root $= 3 \times 10^2 = 300.$

1 mark
300

(2)

(b) Find the value of $\left(\frac{216}{1000}\right)^{-\frac{2}{3}}$

Cube root of 216/1000 = 6/10 = 3/5.

1 mark
Root; reciprocal; square

Power -2 means take the reciprocal and square:

$\left(\frac{5}{3}\right)^2 = \frac{25}{9}$

1 mark
25/9

(2)

(Total for Question 15 is 4 marks)